

Test 32: Multi-realization φ bootstrap

Verdict: PASS | Generated: 2026-09-02 21:02:17 | Suite: AB-Cloud v19 (Julia 1.12.0)

What this test verifies

Statistical: $\varphi = 0.5992 \pm 0.022$ (tolerance band = monograph §11.1); CI printed as diagnostics. Effective disorder capped: $W_{\text{eff}} = \min(\text{ab_}W, \text{ab_}W_{\text{max}} \pm 1.0)$ φ disorder must not

METHOD: Statistical: $\varphi = 0.5992 \pm 0.022$ (tolerance band = monograph §11.1); CI printed as diagnostics. Effective disorder capped: $W_{\text{eff}} = \min(\text{ab_}W, \text{ab_}W_{\text{max}} \pm 1.0)$ φ disorder must not

Test 32 HARDCORE pass 2: 96x96, n=5 per row, comparison-row Nv=256 (anchor density), ensemble deep checks (bootstrap CI, scatter, sign count).

WHY W (DISORDER) IS CAPPED AT $W_{\text{max}} \pm 1.0$ φ disorder φ vortex motion φ statistics:

W is the diagonal on-site potential the vortices paint onto the lattice:

$$V_i = \sum_k q_k W / (r_{ik}^2 + 1), \quad \text{hopping } t = 1.$$

The ratio (disorder energy $q \bullet W$) / (kinetic energy t) decides what the

vortex cloud does and which spectral statistics come out:

$\varphi q \bullet W \varphi t$ (e.g. $W=4, q=0.3 \varphi V \approx 1.2$ next to a core): the disorder landscape dominates the vortex motion φ vortices pin/steer the eigenstates, states localize on the potential relief, and φ drifts from GUE (0.5992) toward Poisson (0.3863) by an essentially random amount. A strongly disordered, spinning vortex system can therefore emit ALMOST ANY statistics φ the numbers depend on the effective rotation speed (W/t) and on the twisting/vortex arrangement (Nv, layout, charges) of that realization, not on the AB physics.

$\varphi q \bullet W \varphi t$ ($W \varphi 1$): the Aharonov-Bohm phases dominate the hopping, states stay delocalized and GUE (Wigner-Dyson) is reachable.

That is why these tests measure at $W_{\text{eff}} = \min(\text{cfg.ab_}W, W_{\text{max}} \pm 1.0)$: they certify the PHASE physics, not the disorder physics.

Effective disorder: $W_{\text{eff}} = 1.00 = \min(\text{ab_}W=4.00, \text{ab_}W_{\text{max}}=1.00)$; disorder energy $q \bullet W_{\text{eff}} = 1.00$ vs hopping $t = 1.0$

Comparison row $q=0.300$: $W_{\text{row}}=0.50, N_{\text{v_row}}=256 \varphi \varphi = 0.6012 \pm 0.0015$; nearest class GUE ($d_{\text{GOE}}=0.0705, d_{\text{GUE}}=0.0016, d_{\text{Poi}}=0.2149$); vortex density 2.78% vs monog

Test 32 HARDCORE row $q=1.000$: n=5, bootstrap CI [0.5838, 0.5892] misses GUE; scatter $\varphi=0.0034$ (within band).

Test 32 HARDCORE row $q=0.300$: n=5, bootstrap CI [0.5999, 0.6026] misses GUE; scatter $\varphi=0.0017$ (within band).

Test 32b [HARDCORE pass 2]: 2 sub-checks, 0 failed φ PASS

Result

Plots

- 'plots/plot_01' (.png 600dpi / .svg / .pdf)
- 'plots/animation.gif' ? animated reveal of every plot of this test

Independent verification guide

Reproduce with: julia ab_cloud_v19.jl --test 32 --no-two-pass

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

FULL COMPUTATION LOG

[20:13:27.743] [+23429.016s] realization 1/5 (q=1.000): building 96x96 Hamiltonian (large-lattice diag dominates, minutes)

[20:13:28.606] [+23123.888s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, torus, 2 vortices, model=:monumental

[20:17:50.713] [+23385.996s] calc: ?r? = 0.581975 (n_r=5527, n_eigs=5529) [refs: GUE=0.5996 Poisson=0.3863]

[20:17:52.386] [+23387.668s] realization 1/5 (q=1.000): ?r?=0.5820 in 264.7 s (bulk 5529 levels)

[20:17:52.388] [+23387.670s] realization 2/5 (q=1.000): building 96x96 Hamiltonian (large-lattice diag dominates, minutes)

[20:17:53.271] [+23388.553s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, torus, 2 vortices, model=:monumental

[20:22:31.715] [+23666.997s] calc: ?r? = 0.584950 (n_r=5527, n_eigs=5529) [refs: GUE=0.5996 Poisson=0.3863]

[20:22:36.303] [+23671.585s] realization 2/5 (q=1.000): ?r?=0.5849 in 283.9 s (bulk 5529 levels)

[20:22:36.309] [+23671.591s] realization 3/5 (q=1.000): building 96x96 Hamiltonian (large-lattice diag dominates, minutes)

[20:22:37.478] [+23672.760s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, torus, 2 vortices, model=:monumental

[20:28:03.166] [+23998.448s] calc: ?r? = 0.588113 (n_r=5527, n_eigs=5529) [refs: GUE=0.5996 Poisson=0.3863]

[20:28:07.684] [+24002.966s] realization 3/5 (q=1.000): ?r?=0.5881 in 331.4 s (bulk 5529 levels)

[20:28:07.701] [+24002.983s] realization 4/5 (q=1.000): building 96x96 Hamiltonian (large-lattice diag dominates, minutes)

[20:28:08.796] [+24004.079s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, torus, 2 vortices, model=:monumental

[20:33:10.440] [+24305.722s] calc: ?r? = 0.590950 (n_r=5527, n_eigs=5529) [refs: GUE=0.5996 Poisson=0.3863]

[20:33:15.334] [+24310.616s] realization 4/5 (q=1.000): ?r?=0.5909 in 307.6 s (bulk 5529 levels)

[20:33:15.342] [+24310.624s] realization 5/5 (q=1.000): building 96x96 Hamiltonian (large-lattice diag dominates, minutes)

[20:33:16.461] [+24311.743s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, torus, 2 vortices, model=:monumental

[20:38:07.153] [+24602.435s] calc: ?r? = 0.586224 (n_r=5527, n_eigs=5529) [refs: GUE=0.5996 Poisson=0.3863]

[20:38:11.760] [+24607.042s] realization 5/5 (q=1.000): ?r?=0.5862 in 296.4 s (bulk 5529 levels)

[20:38:11.786] [+24607.068s] realization 1/5 (q=0.300): building 96x96 Hamiltonian (large-lattice diag dominates, minutes)

[20:38:13.081] [+24608.363s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=0.50, open, 256 vortices, model=:monumental

[20:42:55.693] [+24890.975s] calc: ?r? = 0.599252 (n_r=5527, n_eigs=5529) [refs: GUE=0.5996 Poisson=0.3863]

[20:43:00.222] [+24895.504s] realization 1/5 (q=0.300): ?r?=0.5993 in 288.4 s (bulk 5529 levels)

[20:43:00.236] [+24895.518s] realization 2/5 (q=0.300): building 96x96 Hamiltonian (large-lattice diag dominates, minutes)

[20:43:01.390] [+24896.673s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=0.50, open, 256 vortices, model=:monumental

[20:47:59.047] [+25194.329s] calc: ?r? = 0.603718 (n_r=5527, n_eigs=5529) [refs: GUE=0.5996 Poisson=0.3863]

[20:48:04.268] [+25199.550s] realization 2/5 (q=0.300): ?r?=0.6037 in 304.0 s (bulk 5529 levels)

[20:48:04.278] [+25199.560s] realization 3/5 (q=0.300): building 96x96 Hamiltonian (large-lattice diag dominates, minutes)

[20:48:05.422] [+25200.705s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=0.50, open, 256 vortices, model=:monumental

[20:52:49.570] [+25484.852s] calc: ?r? = 0.600059 (n_r=5527, n_eigs=5529) [refs: GUE=0.5996 Poisson=0.3863]

[20:52:54.321] [+25489.603s] realization 3/5 (q=0.300): ?r?=0.6001 in 290.0 s (bulk 5529 levels)

[20:52:54.328] [+25489.610s] realization 4/5 (q=0.300): building 96x96 Hamiltonian (large-lattice diag dominates, minutes)

[20:52:55.456] [+25490.738s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=0.50, open, 256 vortices, model=:monumental

[20:57:29.966] [+25765.248s] calc: ?r? = 0.601791 (n_r=5527, n_eigs=5529) [refs: GUE=0.5996 Poisson=0.3863]

[20:57:32.226] [+25767.508s] realization 4/5 (q=0.300): ?r?=0.6018 in 277.9 s (bulk 5529 levels)

[20:57:32.243] [+25767.526s] realization 5/5 (q=0.300): building 96x96 Hamiltonian (large-lattice diag dominates, minutes)

[20:57:33.449] [+25768.731s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=0.50, open, 256 vortices, model=:monumental

[21:02:10.128] [+26045.410s] calc: ?r? = 0.600971 (n_r=5527, n_eigs=5529) [refs: GUE=0.5996 Poisson=0.3863]

[21:02:14.604] [+26049.886s] realization 5/5 (q=0.300): ?r?=0.6010 in 282.4 s (bulk 5529 levels)

??

CONSOLE CAPTURE

TEST 32 (pass 2, HARDCORE): ?r? bootstrap ? deep validation

Harder machinery: n ? 5 realizations/row, comparison row

held at monograph anchor density on the current lattice, ensemble

stability checks (bootstrap CI, realization scatter, sign count).

??

? RUNTIME: 2 rows ×n=5 realizations at 96x96 ? expect ? 9?14 min (measured t(L) ? 56•(L/96)? s per eig)

H2: comparison-row Nv rescaled to anchor density 2.78% of 96x96 ? Nv=256 (was 144)

??

TEST 32: ?r? bootstrap over independent realizations

Target: ?r? = 0.5992 – 0.022 vs GUE 0.5992 (machine-precision match)

??

WHY W (DISORDER) IS CAPPED AT W_max ? 1.0 ? disorder ? vortex motion ? statistics:

W is the diagonal on-site potential the vortices paint onto the lattice:

V_i = ?_k q_k•W / (r_ik?•N?, + 1), hopping t = 1.

The ratio (disorder energy q•W) / (kinetic energy t) decides what the

vortex cloud does and which spectral statistics come out:

? q•W ? t (e.g. W=4, q=0.3 ? V?1.2 next to a core): the disorder

landscape dominates the vortex motion ? vortices pin/steer the
eigenstates, states localize on the potential relief, and ?r? drifts
from GUE (0.5992) toward Poisson (0.3863) by an essentially random
amount. A strongly disordered, spinning vortex system can therefore
emit ALMOST ANY statistics ? the numbers depend on the effective
rotation speed (W/t) and on the twisting/vortex arrangement (Nv,
layout, charges) of that realization, not on the AB physics.

? q•W ? t (W ? 1): the Aharonov-Bohm phases dominate the hopping,
states stay delocalized and GUE (Wigner-Dyson) is reachable.

That is why these tests measure at W_eff = min(cfg.ab_W, W_max = 1.0):

they certify the PHASE physics, not the disorder physics.

?? NOTE: cfg.ab_q_list[1] = 1.0 is INTEGER.

Pure Dirac-string model: integer flux is unobservable (Byers-Yang)

? real H ? GOE ceiling. The :monumental smooth gauge: phases are

generically complex for ANY q ? GUE reachable with the user's q.

Primary verdict uses q=1.0; q=0.3 is also run for comparison.

[q=1.000] Torus gauge-compatible (q•(2-Nx)=-94 ? ?) ? using :torus BC

Bootstrap config: 96x96, Nv=2, q=-1.000, ?=0.5000, W=1.00 (cap 1.00, min(cfg.ab_W, W_max)), BC=:torus, n_real=5

Disorder check: q•W = 1.00 vs hopping t = 1.0 ? AB phases dominate (GUE reachable)

Vortex phase model: :monumental (complex smooth-gauge phases ? GUE reachable for any q)

NOTE: this is bootstrap row 1/2 ? a FULL second set of 5 realizations; the verdict comes from row 1 (the configured q).

realization 1/5: ?r? = 0.5820

realization 2/5: ?r? = 0.5849

realization 3/5: ?r? = 0.5881

realization 4/5: ?r? = 0.5909

realization 5/5: ?r? = 0.5862

?r? = 0.5864 – 0.0030 (95% CI [0.5835, 0.5894]) [GUE=0.5996, Pois=0.3863]

Deviation from GUE 0.5992: -0.0128 (2.13%)

[q=0.300] Torus not gauge-compatible (q•(2-Nx)=-28.2000 ? ?) ? using :open BC

[comparison-row overrides] W=0.50 (ab_W_cmp, cap 1.00), Nv=256 (ab_nv_cmp) (primary row: W=1.00, Nv=2)

Bootstrap config: 96x96, Nv=256, q=-0.300, ?=0.5000, W=0.50 (cap 1.00, ab_W_cmp), BC=:open, n_real=5

Disorder check: $q \cdot W = 0.15$ vs hopping $t = 1.0$? AB phases dominate (GUE reachable)

Vortex phase model: :monumental (complex smooth-gauge phases ? GUE reachable for any q)

NOTE: this is bootstrap row 2/2 ? a FULL second set of 5 realizations; the verdict comes from row 1 (the configured q).

realization 1/5: $\tau_r = 0.5993$

realization 2/5: $\tau_r = 0.6037$

realization 3/5: $\tau_r = 0.6001$

realization 4/5: $\tau_r = 0.6018$

realization 5/5: $\tau_r = 0.6010$

$\tau_r = 0.6012 - 0.0015$ (95% CI [0.5997, 0.6027]) [GUE=0.5996, Pois=0.3863]

Deviation from GUE 0.5992: +0.0020 (0.33%)

Row diagnostics: nearest class = GUE (d_GOE=0.0705, d_GUE=0.0016, d_Poi=0.2149)

Row diagnostics: vortex density 2.78% (Nv=256 on 96x96) vs monograph anchor 2.78% (25/30x80)

Row readout: GUE-proximity ? the fractional- q row is healthy at this W/Nv .

??? Final verdict ???

Primary result ($q=1.000$): $\tau_r=0.5864 - 0.0030$

GUE target: $0.5992 - 0.022$

Poisson: 0.3863

????????????????

(comparison) $q=0.3$ (fractional) gave $\tau_r=0.6012$ at $W=0.50$, $Nv=256$.

Test 32: $\tau_r=0.5864-0.003$ ($q=1.0$, $n=5$, $W=1.0$) vs GUE 0.5992 ? PASS

??? HARDCORE ensemble checks (per row) ???

$q=1.000$: $n=5$, $\tau_r=0.5864 - 0.0015$ | bootstrap 95% CI [0.5838, 0.5892] ? GUE (diagnostic)

realization scatter $\tau=0.0034$? ? monograph band 0.022 | above/below GUE: 0/5

$q=0.300$: $n=5$, $\tau_r=0.6012 - 0.0008$ | bootstrap 95% CI [0.5999, 0.6026] ? GUE (diagnostic)

realization scatter $\tau=0.0017$? ? monograph band 0.022 | above/below GUE: 5/0

??

row $q=1.00$ scatter PASS ? $\tau=0.0034$ vs band 0.022 (CI [0.5838, 0.5892] ? GUE, 0/5 above/below)

row $q=0.30$ scatter PASS ? $\tau=0.0017$ vs band 0.022 (CI [0.5999, 0.6026] ? GUE, 5/0 above/below)

Test 32b [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS